

Radical creativity: examining supervisor-subordinate ties on social media

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Abstract

Purpose – Although enterprise social media (ESM) offers new opportunities for developing radical creativity, limited research has been conducted on how to foster such creativity in ESM contexts. Based on the approach-inhibition theory of power (AITP), this study examines how supervisor-subordinate instrumental and expressive ties on ESM promote radical creativity through sense of power. Furthermore, it further investigates how boundary buffering moderates the influence of these supervisor-subordinate ties on sense of power.

Design/methodology/approach – This study adopts a quantitative research design, collecting data from 352 employees through an online survey. The data were analyzed using structural equation modeling (SEM) and fuzzy-set qualitative comparative analysis (fsQCA).

Findings – The findings show that both supervisor-subordinate instrumental and expressive ties on ESM positively enhance employees' sense of power, which in turn promotes ESM-enabled boundary spanning and radical creativity. Boundary buffering strengthens the positive effect of supervisor-subordinate expressive ties on sense of power. Moreover, the fsQCA results identify three distinct configurations that lead to high levels of radical creativity.

Practical implications – Organizations should enhance supervisors' ability to effectively use ESM features to establish diverse connections with subordinates, thereby fostering radical creativity. In addition, organizations should also strengthen employees' sense of power in ESM contexts, thereby motivating them to proactively leverage ESM to access heterogeneous resources and develop radical creativity. Finally, organizations need to support employees in developing boundary-buffering capabilities to fully unlock the empowering and creative potential of ESM.

Originality/value – This study advances research on ESM use and radical creativity by revealing the relationship between supervisor-subordinate ties on ESM and radical creativity. In particular, based on AITP, this study identifies sense of power as the mediating mechanism and boundary buffering as the boundary condition. Furthermore, this study adopts a mixed-method approach to test the theoretical model and the configurational effects of various factors, thereby enriching our understanding of the driving mechanisms of radical creativity in ESM contexts.

Keywords Enterprise social media, Supervisor-subordinate ties, Sense of power, ESM-enabled boundary spanning, Radical creativity

Paper type Research article

1. Introduction

In today's turbulent and competitive environment, organizations need to continuously generate highly novel and useful ideas (i.e. radical creativity) to achieve breakthrough innovations and sustain organizational prosperity (Zhang and Xu, 2024; Yang *et al.*, 2025; Zhu *et al.*, 2025a). Accordingly, identifying effective ways to foster employees' radical creativity has become a central concern for both scholars and practitioners (Zhang and Xu, 2024; Geng *et al.*, 2025). With the rapid advancement of digital technologies, enterprise social media (ESM) has emerged as an open and transparent platform that facilitates communication and collaboration within organizations (Feng *et al.*, 2025; Zhang *et al.*, 2025a; Zheng *et al.*, 2025). ESM enables employees to transcend departmental and hierarchical boundaries (Ma *et al.*, 2025b; Pitafi and Kiani, 2025), allowing them to access diverse knowledge and resources that can drive radical creativity. Many organizations across industries have adopted ESM platforms



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such as Microsoft Yammer and DingTalk to strengthen innovation and competitiveness (Ma *et al.*, 2025b; Sheng *et al.*, 2025). Against this backdrop, it becomes essential to investigate the driving factors of radical creativity in ESM contexts.

Radical creativity refers to employees' ideas that substantially depart from existing practices and drive transformative change, involving high levels of uncertainty and resource investment (Geng *et al.*, 2025; Qi *et al.*, 2025a). Supervisors, who leverage their authority and professional expertise to guide others toward shared goals, are regarded as key drivers of radical creativity by empowering subordinates and providing critical resources (Ren and Song, 2024; Zhang and Li, 2024; Liden *et al.*, 2025). Given recent studies indicating that supervisors influence radical creativity primarily through the attitudes and behaviors they exhibit in ties with subordinates (e.g. supervisor support and leadership styles) (Zhang and Xu, 2024; Song *et al.*, 2025; Zhu *et al.*, 2025a, b), it is necessary to investigate further how supervisors affect radical creativity in ESM contexts, where ESM has transformed the way supervisors and subordinates interact (Budjanovcanin and Yang, 2025; Ollier-Malaterre and Redston, 2025). Although prior research has demonstrated that supervisors play a pivotal role in influencing employees' adoption and effective use of ESM (Hacker *et al.*, 2024; Zhou and Yin, 2025), limited understanding exists regarding how supervisors leverage ESM to foster radical creativity. Compared to traditional communication tools such as email, ESM provides a more transparent and interactive channel that enables supervisors to influence employees by establishing multi-dimensional supervisor-subordinate ties (Ollier-Malaterre and Redston, 2025; Zhou and Yin, 2025). For example, by utilizing various ESM features (e.g. group chats, likes, and greetings), supervisors can flexibly establish instrumental (task-oriented) and expressive (relationship-oriented) ties with subordinates (Wang *et al.*, 2023a; Hacker *et al.*, 2024; Hong *et al.*, 2025), thereby influencing employees by providing different types of information and resources. However, most existing studies have emphasized ESM's affordances and its use in facilitating horizontal communication and collaboration to support employee creativity (Fu *et al.*, 2024; Zhang *et al.*, 2024b; Zhang and Mao, 2025), while offering limited insight into how ESM enables vertical supervisor-subordinate ties to influence the realization of ESM's potential for fostering radical creativity. Without addressing this gap, organizations may fail to design effective leadership interventions that fully leverage ESM's potential to enhance employees' radical creativity, thereby hindering the realization of value from ESM implementations. Therefore, this study proposes the following research question: *What are the underlying mechanisms and boundary conditions under which supervisor-subordinate ties on ESM foster radical creativity?*

Specifically, based on the approach-inhibition theory of power (AITP), this study examines how supervisor-subordinate ties on ESM affect radical creativity. Although these ties provide essential external support for developing radical creativity (Hou *et al.*, 2024; Zhang and Li, 2024), this study argues that their effectiveness may depend on the extent to which they enhance employees' sense of power (i.e. subjective judgments about their ability to control resources and influence others) (Anderson *et al.*, 2012). According to AITP, sense of power individuals derive from social interactions can activate the approach-oriented tendencies (Keltner *et al.*, 2003; Anderson *et al.*, 2012; Kim *et al.*, 2025). This study identifies sense of power as the mediating mechanism linking supervisor-subordinate ties on ESM to radical creativity for two main reasons. First, supervisors are a key source of employees' sense of power, as they convey empowerment and support within these ties (Nguyen *et al.*, 2024; Wu *et al.*, 2025). Compared with previous studies that have emphasized mediators such as creative role identity or cognitive complexity (Xiao *et al.*, 2025; Zhang and Zhang, 2025), sense of power is a social-relational construct that more directly captures employees' psychological responses within supervisor-subordinate ties on ESM, thereby offering a more precise explanation of subsequent behaviors (Anderson *et al.*, 2012; Luo and Chen, 2024). Second, AITP suggests that the approach system triggered by sense of power can accurately predict employees' proactive and risky behaviors (e.g. taking charge behaviors and error risk taking) (Luo *et al.*, 2024; Song and Gu, 2024b), which align closely with the uncertainty and boldness

inherent in radical creativity (Ren and Song, 2024; Geng *et al.*, 2025). This indicates that sense of power may serve as a motivational mechanism through which employees use ESM to span boundaries and engage in creative exploration. Although previous research emphasizes the empowering capabilities of ESM affordances (Sun *et al.*, 2024; Khan *et al.*, 2025), there has been limited focus on how supervisors leverage ESM to enhance employees' sense of power and foster radical creativity. Therefore, this study investigates the mediating role of sense of power in the relationship between supervisor-subordinate ties on ESM and radical creativity, offering both theoretical insight and practical guidance for developing leadership strategies that foster radical creativity in ESM contexts.

Furthermore, this study examines the boundary conditions under which supervisor-subordinate ties influence employees' sense of power in ESM contexts, highlighting how employees can effectively derive sense of power to foster radical creativity. According to AITP, employees' sense of power depends on their subjective judgments about their relative controllability within social interactions (Anderson *et al.*, 2012; Wang *et al.*, 2025). Although frequent supervisor-subordinate ties on ESM improve employees' access to resources and support, constant communication can also cause information overload and interruptions (Pitafi *et al.*, 2025a; Sun *et al.*, 2025), which may reduce their perceived autonomy in managing work. Therefore, an effective way for employees to maintain their sense of power may be to engage in boundary buffering (i.e. avoiding excessive interference and demands from the ESM) (Zhang *et al.*, 2022; Lei *et al.*, 2024). Boundary buffering allows employees to retain autonomy during online interactions by regulating when and how they engage with others (Zhang *et al.*, 2022; Xi *et al.*, 2025). For example, employees can use silent mode or permission settings on ESM to avoid unnecessary distractions. Although previous studies have shown that boundary buffering can mitigate job stress and enhance mindfulness (Zhang *et al.*, 2022; Rodriguez-Escudero *et al.*, 2025), its role in shaping employees' sense of power within ESM-based social interactions remains underexplored. Combining AITP with boundary buffering, this study argues that employees with boundary buffering capabilities are more effective at obtaining sense of power from supervisor-subordinate ties on ESM. Therefore, this study explores the moderating role of boundary buffering to provide practical insights into how organizations can balance open communication with employee autonomy, thereby optimizing ESM use to foster radical creativity.

Based on the above discussion, this study draws on AITP to develop a theoretical framework that explores how supervisor-subordinate instrumental and expressive ties on ESM influence radical creativity, while also examining the moderating role of boundary buffering. This study makes several important contributions. First, it advances research on ESM use and radical creativity by clarifying the relationship between supervisor-subordinate ties on ESM and radical creativity. Second, it deepens our understanding of the mediating role of employees' sense of power, illustrating the psychological mechanisms through which different types of supervisor-subordinate ties influence radical creativity in ESM contexts. Third, it highlights the moderating role of boundary buffering, revealing the conditions under which these ties most effectively enhance employees' sense of power. Finally, this study provides guidance for how organizations can utilize ESM to foster radical creativity in ESM contexts.

2. Theoretical background

2.1 ESM use

ESM functions as a digital platform for internal communication and collaboration, enabling employees to post, share, and exchange user-generated content (Pitafi *et al.*, 2025b; Sheng *et al.*, 2025). ESM affordances, such as visibility, persistence, editability, and associability, enhance the transparency, accessibility, and traceability of information, thereby improving resource integration and interpersonal connection (Sun *et al.*, 2024; Pitafi, 2025). By providing an open and transparent environment, ESM helps break down structural barriers caused by task specialization, departmental divisions, and organizational hierarchies (Zhang *et al.*, 2022;

[Ma et al., 2025b](#)). It enables employees to engage in boundary spanning by forming social ties within and across teams to access knowledge, resources, and support ([Hong et al., 2025](#)). Through ESM-enabled boundary spanning, employees can overcome temporal and spatial constraints to obtain diverse and heterogeneous knowledge essential for innovation ([Zhang and Li, 2024](#); [Ma et al., 2025b](#)). Furthermore, ESM provides customizable features (e.g. mute settings and online status controls) that facilitate boundary buffering, enabling employees to exercise autonomy and manage unwanted interruptions while pursuing work goals ([Zhang et al., 2023](#); [Lei et al., 2024](#)). These recent studies indicate that ESM supports employees in effectively accessing and integrating resources through flexible boundary management mechanisms (i.e. boundary spanning and boundary buffering) ([Zhang et al., 2022](#); [Ma et al., 2025b](#)), thereby providing a crucial foundation for the development of radical creativity ([Bonaccorsi and Lyytinen, 2025](#)).

A review of the existing literature on ESM use and creativity reveals two primary research streams. The first focuses on how ESM's technological features promote employee creativity, drawing on theories such as communication visibility and affordance theory ([Fu et al., 2024](#); [Shen and Wang, 2024](#); [Ma et al., 2025a](#); [Shao et al., 2025b](#)). For example, [Fu et al. \(2024\)](#) found that visibility promotes employee creativity by increasing experience and network recognition. The second stream examines how different patterns of ESM use influence creativity, drawing on theories such as conservation of resource theory, task-technology fit, and the job demands-resources model ([Zhu et al., 2024](#); [Zhang et al., 2025a](#)). For example, [Zhu et al. \(2024\)](#) found that work-related and social-related ESM use can foster creativity by promoting employees' psychological meaningfulness, availability, and safety. Although recent studies have enriched our understanding of the relationship between ESM use and employee creativity, several research gaps remain. First, recent studies on ESM use often treat employee creativity as a unidimensional construct, which constrains our understanding of how ESM use supports radical creativity. Prior studies indicate that the same antecedents can have markedly different effects on these two forms of creativity ([Qi et al., 2025a](#); [Zhu et al., 2025a](#)). As organizations increasingly depend on original and disruptive ideas to sustain competitiveness in dynamic environments ([Geng et al., 2025](#); [Zheng et al., 2025](#)), it is necessary to explore further how to leverage ESM to foster radical creativity. Second, although research has shown that ESM possesses strong functional potential to enhance creativity, it offers limited insight into how supervisors can leverage it effectively. ESM has become a vital channel for supervisor-subordinate communication, offering supervisors new avenues to guide and influence employee behavior ([Budjanovcanin and Yang, 2025](#); [Ollier-Malaterre and Redston, 2025](#)). Although recent studies suggest that leadership plays a crucial role in promoting ESM adoption and effective use ([Hacker et al., 2024](#)), it remains unclear how supervisors influence radical creativity in ESM contexts. Addressing this gap can provide both theoretical and practical insights, helping managers design interventions that fully realize ESM's potential to drive radical creativity.

2.2 Supervisor-subordinate ties on ESM

In the workplace, supervisor-subordinate ties form a central vertical relationship that continuously shapes employees' attitudes and behavior ([Hou et al., 2024](#); [Williams et al., 2025](#)). These ties not only reflect formal hierarchical relationships in which supervisors deliver strategic goals, clarify behavioral norms, and provide legitimacy to support employees' work ([Gu et al., 2024](#); [Zhu et al., 2025c](#)), but also encompass functional exchanges in which supervisors provide guidance, experience, and resources to improve employees' task efficiency ([Lee et al., 2025](#)). High-quality supervisor-subordinate ties facilitate mutual understanding and resource flow between supervisors and employees ([Hamstra et al., 2024](#); [Williams et al., 2025](#)), serving as a primary channel for enhancing task performance, well-being, and psychological empowerment ([Liu et al., 2024](#); [Borut et al., 2025](#); [Lee et al., 2025](#)). Although recent studies have highlighted the importance of supervisor support and leadership

styles (e.g. participative leadership, secure-base leadership) in promoting radical creativity (Kumar *et al.*, 2024; Zhang and Li, 2024; Song *et al.*, 2025; Zhu *et al.*, 2025a), they overlook that these leadership behaviors and styles are primarily enacted and perceived through supervisor-subordinate ties (Williams *et al.*, 2025). Williams *et al.* (2025) point out that supervisors may establish different types of ties with subordinates, underscoring the need to examine how the nature of these ties influences employee outcomes such as radical creativity.

With the digital transformation of the workplace, ESM has become a key channel for communication and collaboration across organizational hierarchies (Budjanovcanin and Yang, 2025; Hong *et al.*, 2025; Zhou and Yin, 2025). Beyond facilitating horizontal communication among employees, ESM also enhances vertical interactions between supervisors and subordinates (Budjanovcanin and Yang, 2025; Ollier-Malaterre and Redston, 2025). ESM enables supervisors to engage more directly and frequently with employees, allowing them to monitor progress, provide timely feedback, and offer guidance (Kim and Yoon, 2025; Ollier-Malaterre and Redston, 2025). At the same time, ESM helps supervisors identify employees' emotional and social support needs through features such as status updates, comments, and reactions, enabling more empathetic and personalized communication (Kim and Yoon, 2025; Ollier-Malaterre and Redston, 2025). Therefore, ESM reduces traditional hierarchical barriers and supports the development of more direct and diverse supervisor-subordinate ties (Budjanovcanin and Yang, 2025; Ollier-Malaterre and Redston, 2025). Based on the content of communication, supervisor-subordinate ties on ESM can be classified into instrumental ties and expressive ties (Hong *et al.*, 2025; Pitafi and Kiani, 2025). Specifically, instrumental ties refer to task-related interactions in which supervisors communicate with employees about formal work matters, primarily by offering advice, knowledge, and experience to support task completion (Pitafi and Kiani, 2025; Zhang *et al.*, 2025b). For example, supervisors can utilize ESM features, such as activity updates and task tracking, to monitor employees' progress, provide timely feedback, and coordinate resources to facilitate goal achievement (Ollier-Malaterre and Redston, 2025). In contrast, expressive ties involve emotional and social support, addressing employees' needs for care, recognition, and a sense of belonging (Pitafi and Kiani, 2025; Zhang *et al.*, 2025b). For example, supervisors may express concern and encouragement by liking posts, commenting, or sending holiday greetings through ESM, thereby fostering emotional bonds and psychological safety in routine communication. Although recent qualitative studies have shown that ESM transforms supervisor-subordinate communication and enables supervisors to influence employee attitudes and behaviors through more direct and diverse ties (Hacker *et al.*, 2024; Budjanovcanin and Yang, 2025; Ollier-Malaterre and Redston, 2025), empirical evidence remains limited on how different types of supervisor-subordinate ties influence employees' radical creativity.

2.3 Approach-inhibition theory of power

According to AITP, sense of power is a social and relational construct that reflects an individual's subjective perception of their ability to influence others' behavior (Anderson *et al.*, 2012; Luo and Chen, 2024). AITP posits that this sense of power is not determined solely by formal organizational position or hierarchy but also by individuals' perceived control over key resources such as knowledge, expertise, and social capital within their interactions (Ni and Zheng, 2024; Song and Gu, 2024b). Compared to objective structural power, the subjective sense of power is a stronger predictor of employee behavior (Ni and Zheng, 2024; Wang *et al.*, 2025). The widespread adoption of digital technologies (e.g. ESM) has led to flatter organizational structures, expanded social networks, and increased information visibility, broadening the channels through which employees derive their sense of power (Sun *et al.*, 2024; Khan *et al.*, 2025). Previous studies have shown that ESM affordances support employees in more actively accessing resources, expressing opinions, and establishing influence (Sun *et al.*, 2024; Khan *et al.*, 2025). However, while prior research emphasizes

ESM's potential to empower employees through its technological features, less is known about how this potential is realized in everyday work practices. Given that ESM is a socially embedded and malleable tool and sense of power is fundamentally constructed through social interactions (Hacker *et al.*, 2024; Wu *et al.*, 2025), its empowering effects may depend on how it is enacted through daily interactions. As supervisors play a central role in shaping employees' sense of power (Nguyen *et al.*, 2024; Wu *et al.*, 2025), it is essential to investigate how supervisor-subordinate ties on ESM influence employees' sense of power.

In organizational contexts, individuals at different hierarchical levels experience varying degrees of sense of power shaped by their daily interactions and situational dynamics (Kim *et al.*, 2025). According to AITP, different levels of sense of power trigger distinct emotional, cognitive, and behavioral responses (Song and Gu, 2024a; Funk, 2025). Specifically, a high sense of power activates the approach system, fostering optimism toward risk and motivating proactive exploration of opportunities and rewards (Wang *et al.*, 2024; Kim *et al.*, 2025). In contrast, a low sense of power activates the inhibition system, leading to negative emotions, risk aversion, and behavioral restraint (Magni *et al.*, 2022; Wang *et al.*, 2025). AITP thus provides a valuable theoretical framework for understanding how the sense of power influences individual behavior within organizations. As a malleable psychological state, sense of power not only strongly predicts employee behavior but also provides a theoretical foundation for developing interventions that enhance positive and proactive work behaviors (Ni and Zheng, 2024; Kim *et al.*, 2025). Strengthening employees' sense of power to encourage approach-oriented behaviors has become an important concern for both researchers and practitioners (Luo and Chen, 2024; Luo *et al.*, 2024). Therefore, based on AITP, this study develops a theoretical model to explore how supervisor-subordinate ties on ESM foster boundary spanning and radical creativity through employees' sense of power.

3. Hypothesis development

3.1 Supervisor-subordinate social ties on ESM and sense of power

In the digital workplace, supervisor-subordinate expressive ties on ESM refer to the relationship-oriented social connections formed through informal and nonwork interactions between supervisors and subordinates (Wang *et al.*, 2023a). According to AITP, sense of power originates from the group to which an individual belongs and is shaped by various contextual factors, including social dynamics and group interactions (Keltner *et al.*, 2003; Ni and Zheng, 2024). Within ESM environments, expressive ties create a context that promotes more equal interactions between supervisors and subordinates, reducing power distance and enhancing employees' autonomy (Ma *et al.*, 2025b). These ties foster trust and psychological safety, allowing employees to freely express their ideas and overcome hierarchical constraints without fear of negative consequences (Seppala *et al.*, 2011; Khan *et al.*, 2023). The mutual respect and care communicated through ESM interactions strengthen employees' sense of belonging and psychological ownership of their roles within the organization (Hou *et al.*, 2024). Furthermore, within Chinese workplace culture, such informal ties often result in tangible rewards and promotions, further enhancing employees' perceived influence within the organization (Zhou *et al.*, 2019). Therefore, expressive ties on ESM create a favorable group dynamic that enhances employees' sense of power by reshaping hierarchical relationships and promoting trust-based communication. Based on the above discussion, this study hypothesizes:

H1. Supervisor-subordinate expressive ties on ESM positively affect sense of power.

Supervisor-subordinate instrumental ties on ESM refer to task-oriented social connections formed through work-related and formal interactions between supervisors and subordinates (Wang *et al.*, 2023a). Drawing on AITP, employees' sense of power is also shaped by their access to valuable resources and opportunities for participation in group decision-making (Keltner *et al.*, 2003; Ni and Zheng, 2024). Instrumental ties on ESM facilitate efficient

communication, enabling employees to access key resources, receive timely feedback, and gain recognition for their contributions (Wang *et al.*, 2022a). These ties foster high-quality leader-member exchanges characterized by mutual respect and trust (Wang *et al.*, 2022d). Furthermore, ESM reduces information asymmetry by increasing transparency and accessibility to organizational decision-making processes (Deng, 2023; Yu *et al.*, 2024). Employees can directly participate in decision-making discussions and offer suggestions to supervisors, fostering a sense of involvement and influence within the group. Through these mechanisms, instrumental ties on ESM allow employees to feel integrated into the organizational group dynamic and enhance their sense of power by providing them with greater control over resources and decision-making opportunities. Based on the above discussion, this study hypothesizes:

H2. Supervisor-subordinate instrumental ties on ESM positively affect sense of power.

3.2 Sense of power and ESM-enabled boundary spanning

Sense of power refers to an individual's psychological perception of his or her ability to control and influence others within their work team (Anderson *et al.*, 2012). This perception stems from one's access to and control over valuable resources (Yu and Blader, 2020; Korman *et al.*, 2022). In the digital workplace, employees can draw on information, knowledge, skills, and other resources available through online platforms, which further enhance their sense of power (Seppala *et al.*, 2011). Thus, employees with a high sense of power possess greater instrumental resources and confidence in leveraging them effectively. According to AITP, individuals with a high sense of power are more likely to activate an approach behavior system, characterized by proactive, goal-oriented actions, risk-taking, and creative engagement (Keltner *et al.*, 2003). Therefore, sense of power makes employees focused on achieving their goals, more confident in their behavior, and optimistic about various risk events, positively affecting their voice, knowledge sharing, and job performance (Lin *et al.*, 2019; Young-Hyman and Kleinbaum, 2020; Luqman *et al.*, 2022; Issac *et al.*, 2023). Furthermore, employees with a high sense of power tend to think more abstractly and strategically (Smith and Trope, 2006). This cognitive orientation provides both the ability and motivation to engage in boundary spanning, enabling them to acquire valuable resources from diverse external sources (Guinote, 2017). Based on the above discussion, this study proposes the following hypothesis:

H3. Sense of power positively affects ESM-enabled boundary spanning.

3.3 Sense of power and radical creativity

Radical creativity refers to generating highly novel ideas that meaningfully differ from existing practices, which requires employees with stronger motivation and the ability to cope with the complexities and challenges of realizing radical creativity (Madjar *et al.*, 2011; Jiang *et al.*, 2023). According to AITP, individuals with a high sense of power are more likely to engage in approach-oriented behaviors, pursuing opportunities and rewards with confidence and initiative (Ni and Zheng, 2024). Therefore, this study argues that the sense of power positively influences radical creativity. First, employees with a high sense of power are less constrained by external evaluations and exhibit greater confidence in their own ideas, which fosters unconventional thinking and the generation of novel solutions (Zhou and He, 2020). Moreover, a high sense of power also enhances employees' influence within teams (Guinote, 2017), allowing them to access and mobilize diverse resources across organizational boundaries. This increased resource integration strengthens their ability to combine heterogeneous knowledge and drive radical creativity. Finally, individuals with a high sense of power tend to focus on potential gains and demonstrate greater risk tolerance (Magni *et al.*, 2022; Luo *et al.*, 2024), making them more willing to embrace uncertainty and proactively

explore uncharted domains. Based on the above discussion, this study proposes the following hypothesis:

H4. Sense of power positively affects radical creativity.

3.4 Boundary spanning and radical creativity

Employees' boundary spanning behavior enables them to overcome cognitive and knowledge inertia, allowing them to establish more connections with external stakeholders and access more heterogeneous resources (Fang *et al.*, 2021; Wang *et al.*, 2022b), thereby providing a basis for radical creativity. The visibility and persistence of ESM affordances further enhance this process by allowing employees to access organizational knowledge and improve their meta-knowledge accuracy (Van Osch and Steinfeld, 2016, 2018; Zhang *et al.*, 2022). Through ESM-enabled boundary spanning, employees can obtain nonredundant external knowledge, broaden their knowledge base, and develop diverse perspectives (Mell *et al.*, 2022). ESM also reduces hierarchical barriers, enabling informal communication with external members that encourages divergent thinking and creative exploration (Ali *et al.*, 2024). Moreover, boundary spanning behaviors increase employees' network centrality and enhance their influence within teams (Liu *et al.*, 2018; Ze *et al.*, 2018), allowing them to enact informal leadership roles that facilitate idea implementation. By acquiring heterogeneous knowledge and expanding their influence through boundary spanning, employees strengthen both the generation and realization of radical ideas (Wang *et al.*, 2022c). Based on the above discussion, this study proposes the following hypothesis:

H5. Boundary spanning positively affects radical creativity.

3.5 The moderating role of boundary buffering

Boundary buffering refers to the entity's efforts to process external requests to protect it from the external environment (Wang *et al.*, 2023b). Prior research has primarily examined boundary buffering within team boundary management, showing that it can enhance team performance by fostering psychological safety among members (Faraj and Yan, 2009). For example, teams can protect their resources and weaken external pressures by buffering activities such as refusing unreasonable requests (Leicht-Deobald *et al.*, 2022). However, the implication of ESM has blurred traditional departmental and team boundaries, exposing employees to a constant flow of information and requests (Ma *et al.*, 2025b). Therefore, scholars emphasize the importance of employees actively engaging in boundary buffering behaviors to protect themselves from unnecessary interruptions and excessive demands in ESM contexts (Zhang *et al.*, 2022).

This study argues that boundary buffering enables employees to derive a stronger sense of power from supervisor-subordinate ties on ESM. According to AITP, individuals' sense of power is shaped by their perceived level of relative control within social interactions (Anderson *et al.*, 2012; Wang *et al.*, 2025). Boundary buffering allows employees to regulate the frequency and content of their interactions with supervisors, enhancing their perceived control and autonomy. For example, employees may use ESM features such as status indicators or automated replies to signal that messages will be addressed within a set timeframe rather than immediately. This perceived control enables employees to be more autonomous in utilizing the resources provided by their supervisors to achieve their goals, which in turn promotes a sense of power. In addition, boundary buffering reduces anxiety associated with constant visibility to supervisors, enhancing employees' psychological safety in supervisor-supervisor ties. For example, employees can use privacy settings on ESM to selectively share personal content or limit exposure. Previous research suggests that boundary buffering enhances psychological safety by maintaining appropriate relational boundaries (Faraj and Yan, 2009). By maintaining an appropriate psychological distance, employees can

maximize the positive effects of instrumental and expressive ties on their sense of power. In contrast, employees who lack effective boundary buffers may face information overload and excessive demands in supervisor-subordinate ties, making it difficult to derive a sense of power from these ties. Based on the above discussion, this study proposes the following hypotheses:

- H6a.* Boundary buffering plays a positive moderating role between supervisor-subordinate expressive ties on ESM and sense of power.
- H6b.* Boundary buffering plays a positive moderating role between supervisor-subordinate instrumental ties on ESM and sense of power.

Based on the above discussions, this study proposes a research model, as shown in [Figure 1](#).

4. Research methods

This study employed a quantitative cross-sectional research design using a survey-based approach to examine the drivers of radical creativity in ESM contexts. To ensure a comprehensive empirical analysis, two complementary analytical methods were applied: partial least squares structural equation modeling (PLS-SEM) and fuzzy-set qualitative comparative analysis (fsQCA). Combining these methods enabled the examination of both symmetric (net effects) and asymmetric (multiple causal pathways) relationships among variables, thereby strengthening the theoretical depth and empirical robustness of the findings.

4.1 Measures

The measurement items for each construct were adapted from well-established scales, as summarized in [Table 1](#). Supervisor-subordinate expressive and instrumental ties on ESM were measured using four items adapted from [Wang et al. \(2023a\)](#), sense of power was measured using eight items adapted from [Anderson et al. \(2012\)](#), ESM-enabled boundary spanning was measured using four items adapted from [Mell et al. \(2022\)](#), radical creativity was measured using four items adapted from [Jiang et al. \(2023\)](#) and [Jauss and Randel \(2014\)](#), and boundary buffering was measured using three items adapted from [Zhang et al. \(2022\)](#). Because the data were collected in China, the questionnaire was developed using the English-Chinese back-translation method to ensure linguistic and conceptual equivalence. Two professors and four doctoral students with expertise in information systems research reviewed the instrument to ensure its clarity and validity. All items were rated on a seven-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree).

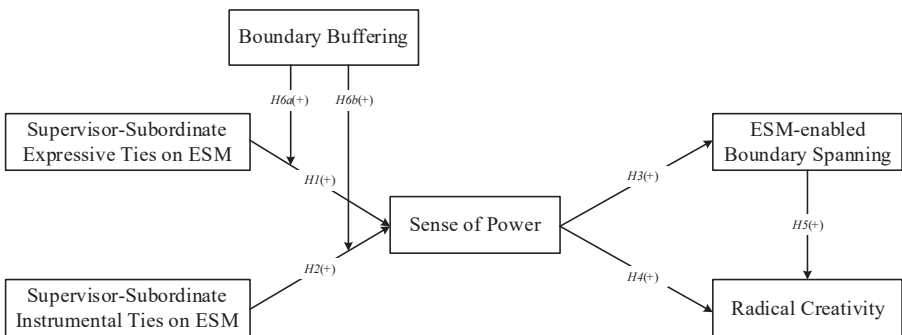


Figure 1. Research model

Table 1. Measurement items

Constructs	Items measured
Supervisor-Subordinate Expressive Ties on ESM (SSET) (Wang <i>et al.</i> , 2023a)	SSET1: Through ESM, I am well acquainted personally with my supervisor SSET2: My supervisor frequently engages in conversations with me about non-work-related topics through ESM SSET3: My supervisor seeks my advice on personal matters through ESM SSET4: My supervisor and I have established meaningful friendships with each other through ESM
Supervisor-Subordinate Instrumental Ties on ESM (SSIT) (Wang <i>et al.</i> , 2023a)	SSIT1: Through ESM, my supervisor frequently shares information with me for coordination, control, planning, or evaluation purposes SSIT2: I often receive technical support and guidance from my supervisor through ESM SSIT3: The interactions I have with my supervisor through ESM are crucial for my job performance SSIT4: My supervisor and I frequently exchange work-related advice and suggestions through ESM
Sense of Power (SP) (Anderson <i>et al.</i> , 2012)	SP1: In work teams, I can get him/her/them to listen to what I say SP2: In work teams, my wishes do not carry much weight. (r) SP3: In work teams, I can get him/her/them to do what I want SP4: In work teams, even if I voice them, my views have little sway SP5: In work teams, I think I have a great deal of power SP6: In work teams, my ideas and opinions are often ignored. (r) SP7: In work teams, even when I try, I am not able to get my way. (r) SP8: In work teams, if I want to, I get to make the decisions
ESM-enabled Boundary Spanning (BS) (Mell <i>et al.</i> , 2022)	BS1: I use ESM to solicit information and resources from external channels BS2: I make use of my relationships with external colleagues through ESM to support my work BS3: I actively use ESM to solicit external work resources BS4: My work depends upon information and resources actively solicited through ESM, that is, information and resources beyond what comes through official channels
Radical creativity (RC) (Jaussi and Randel, 2014; Jiang <i>et al.</i> , 2023)	RC1: In my work, I often have novel and innovative ideas RC2: I often display originality in my work RC3: I can identify job opportunities in a new project operation/new product development/new workflow RC4: I have confidence in my ability to solve problems creatively
Boundary Buffering (BB) (Zhang <i>et al.</i> , 2022)	BB1: I take measures (e.g. muting nonwork-related notifications) to ensure that I can work free of ESM interference BB2: I prevent myself from receiving overloaded information or requests on ESM BB3: I inform others on ESM that I'm concentrating on my work (e.g. by setting a "busy" mode on ESM)

4.2 Data collection

This study focused on China’s software industry as the research context for two main reasons. First, ESM such as DingTalk and WeCom have achieved widespread adoption in China, with particularly intensive use in the software services sector due to its highly digitalized and

collaboration-intensive nature. Second, employees in this industry are typically engaged in knowledge-intensive and creative work, such as product development and system design, where radical creativity plays a central role in both individual performance and organizational innovation. Because of their extensive use of ESM and the importance of radical creativity in their daily work, employees in the software industry are a theoretically suitable and practically relevant group for studying how supervisor-subordinate ties influence radical creativity in ESM contexts.

This study adopted an online questionnaire to collect data. To ensure both the efficiency and quality of data collection, we utilized a professional Chinese survey platform, which is similar to Amazon's MTurk and has been widely adopted in academic research (Shao *et al.*, 2025a; Yu *et al.*, 2025). Participants were recruited via the platform's sample service, and monetary incentives were provided to those who completed the survey. Only individuals who had experience using ESM and whose work involved innovation-related tasks were eligible to complete the full questionnaire. Before beginning the main survey, participants were informed of the study's privacy policy and instructed to respond based on their recent work experiences. The data collection period lasted for one week, during which 378 responses were received. After excluding cases based on trap questions and answering time, 352 valid responses were retained for subsequent data analysis. Specifically, this study included ESM experience, team size, and team tenure as control variables. The demographic characteristics of the respondents are presented in Table 2.

5. Analysis and results

This study employed a mixed-method approach, combining PLS-SEM and fsQCA to comprehensively analyze the proposed research model. First, SmartPLS 3.0 was used to conduct the PLS-SEM analysis. This method was chosen for two main reasons. First, this study aimed to theoretically integrate and examine the key factors that lead to radical

Table 2. Descriptive statistics of respondent characteristics

Demographic variable		Size	%
Gender	Male	121	34.38
	Female	231	65.63
Age	<25 years old	77	21.88
	26–35 years old	197	55.97
	36–45 years old	65	18.47
	>46 years old	13	3.69
Education	Senior middle school or below	11	3.13
	Junior college	289	82.10
	Bachelor's degree	49	13.92
	Master's degree or above	3	0.85
ESM experience	<= 1 year	26	7.39
	1–3 years	160	45.45
	3–5 years	114	32.39
	>= 5 years	52	14.77
Team tenure	<= 6 months	33	9.38
	7–12 months	88	25.00
	13–24 months	90	25.57
	>= 24 months	141	40.06
Team size	<= 5	23	6.53
	6–10	70	19.89
	11–20	104	29.55
	21–30	58	16.48
	>30	97	27.56

creativity. PLS-SEM is particularly suitable for building and testing exploratory theoretical models (Hair *et al.*, 2012). Second, PLS-SEM is robust to complex model structures, small to medium sample sizes, and non-normal data distributions (Hair *et al.*, 2012). It also effectively addresses multicollinearity and allows for precise estimation of complex relationships among latent variables, including mediation and moderation effects.

This study first assessed the measurement model to evaluate the reliability and validity of the constructs, followed by an assessment of the structural model to test the hypothesized relationships. Although PLS-SEM provides valuable insights into the net effects among variables, it assumes linear and symmetric causal relationships across the sample. This limitation constrains its ability to capture how different combinations of antecedent conditions may jointly lead to a particular outcome (Pappas and Woodside, 2021). In practice, radical creativity within organizations often emerges through non-linear, asymmetric, and multiple pathways, requiring complementary analytical approaches to fully understand its complexity.

This study further employed fsQCA 4.0 to examine the configurational relationships among variables. Grounded in Boolean logic and fuzzy-set theory, fsQCA emphasizes causal complexity and configurational reasoning. It identifies multiple combinations of causal conditions that can lead to the same outcome (Qi *et al.*, 2025b). This approach is particularly suitable for uncovering how interactions among various factors jointly contribute to radical creativity, an aspect that traditional symmetric methods may overlook.

In summary, combining PLS-SEM and fsQCA provides a complementary methodological approach (Kang *et al.*, 2024; Qi *et al.*, 2025b). PLS-SEM identifies the average net effects among variables, whereas fsQCA uncovers multiple causal pathways based on different configurations of conditions. Integrating these methods enhances the robustness of the analysis and offers deeper theoretical insights into the mechanisms that drive radical creativity in ESM contexts.

5.1 Measurement model

The reliability and validity of the measurement model were assessed. As shown in Table 3, Cronbach’s alpha values for all constructs ranged from 0.781 to 0.919, and composite reliability values ranged from 0.864 to 0.934, both exceeding the recommended threshold of 0.70, indicating satisfactory internal consistency (Hair *et al.*, 2009). As shown in Tables 3 and 4, the average variance extracted (AVE) for each construct was above 0.60, and all factor loadings exceeded 0.70, demonstrating good convergent validity. In addition, the square root of each construct’s AVE was greater than its inter-construct correlations, supporting discriminant validity. Furthermore, as shown in Table 5, all heterotrait-monotrait (HTMT) ratios were below 0.85, providing additional evidence of discriminant validity (Henseler *et al.*, 2015).

To minimize potential bias from self-reported data, this study conducted tests for common method bias. First, the Harman single-factor test was applied to 27 items across six variables. The results showed that the first factor accounted for 45.181% of the variance, which is below

Table 3. Descriptive statistics and inter-construct correlations

Constructs	Cronbach’s alpha	CR	AVE	SSET	SSIT	SP	BS	RC	BB
SSET	0.854	0.902	0.698	0.835					
SSIT	0.793	0.864	0.614	0.545	0.784				
SP	0.919	0.934	0.640	0.695	0.539	0.800			
BS	0.825	0.884	0.655	0.619	0.617	0.642	0.810		
RC	0.855	0.902	0.697	0.604	0.630	0.734	0.664	0.835	
BB	0.781	0.873	0.696	0.545	0.489	0.498	0.569	0.527	0.834

Table 4. Cross loadings

Item	SSET	SSIT	SP	BS	RC	BB
SSET1	0.766	0.524	0.581	0.464	0.501	0.423
SSET2	0.858	0.373	0.559	0.492	0.445	0.419
SSET3	0.848	0.372	0.565	0.522	0.480	0.451
SSET4	0.864	0.538	0.612	0.585	0.584	0.521
SSIT1	0.382	0.770	0.421	0.445	0.485	0.321
SSIT2	0.331	0.767	0.340	0.429	0.464	0.323
SSIT3	0.567	0.823	0.513	0.545	0.513	0.461
SSIT4	0.381	0.773	0.381	0.501	0.513	0.408
SP1	0.532	0.358	0.789	0.467	0.493	0.381
SP2	0.576	0.470	0.845	0.545	0.582	0.356
SP3	0.599	0.412	0.824	0.476	0.527	0.409
SP4	0.472	0.534	0.750	0.549	0.659	0.394
SP5	0.582	0.375	0.833	0.483	0.549	0.374
SP6	0.512	0.446	0.732	0.538	0.606	0.378
SP7	0.572	0.471	0.803	0.535	0.675	0.466
SP8	0.601	0.360	0.818	0.498	0.580	0.417
BS1	0.521	0.479	0.510	0.817	0.539	0.474
BS2	0.532	0.489	0.523	0.799	0.539	0.469
BS3	0.479	0.476	0.511	0.807	0.495	0.440
BS4	0.473	0.551	0.533	0.815	0.574	0.461
RC1	0.515	0.504	0.598	0.552	0.837	0.442
RC2	0.528	0.525	0.614	0.547	0.822	0.418
RC3	0.503	0.525	0.646	0.574	0.856	0.460
RC4	0.471	0.551	0.591	0.544	0.824	0.438
BB1	0.436	0.451	0.398	0.530	0.477	0.833
BB2	0.467	0.411	0.419	0.449	0.398	0.846
BB3	0.460	0.365	0.427	0.449	0.445	0.823

Management
Decision**Table 5.** HTMT ratio

Items	SSET	SSIT	SP	BS	RC	BB
SSET						
SSIT	0.638					
SP	0.783	0.614				
BS	0.736	0.754	0.734			
RC	0.705	0.764	0.823	0.789		
BB	0.665	0.613	0.585	0.710	0.645	

the 50% threshold. Second, a full collinearity assessment was performed, and the variance inflation factor (VIF) values ranged from 1.510 to 2.676, well below the recommended threshold of 3.3 (Kock, 2015). These results indicate that common method bias was not a significant concern in this study.

5.2 Structural model

The study used a bootstrap approach with 5,000 resamples to test the hypotheses; the use of 5,000 samples for the bootstrap analysis was consistent with common practice and standards in the literature. The SRMR value was 0.072, below the threshold of 0.08 (Henseler *et al.*, 2016). Therefore, the proposed model had a good structural model fit. The results of the structural model analysis were shown in Figure 2. Taking gender, age, education, ESM experience, team tenure, and team size as control variables, the results showed that supervisor-subordinate

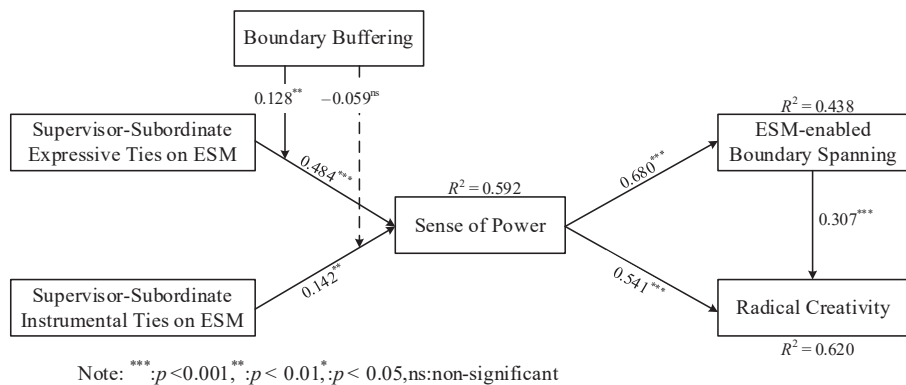


Figure 2. Results of the structural model analysis

expressive ties positively affected sense of power ($\beta = 0.484, t = 8.423$) and supervisor-subordinate instrumental ties positively affected sense of power ($\beta = 0.142, t = 2.733$); thus, H1 and H2 were supported. The sense of power positively influenced ESM-enabled boundary spanning ($\beta = 0.680, t = 17.527$) and radical creativity ($\beta = 0.541, t = 9.190$), thus supporting H3 and H4. ESM-enabled boundary spanning also positively affected radical creativity ($\beta = 0.307, t = 4.714$), thus supporting H5.

5.3 Mediating test

Mediating effects were measured using the bootstrapping method in SmartPLS 3.0 (Nitzl et al., 2016), where the effect of the mediator variable is supported when the 95% confidence interval for the indirect effect does not include zero. As shown in Table 6, sense of power played a mediating role between supervisor-subordinate ties and radical creativity, as well as between supervisor-subordinate ties and ESM-enabled boundary spanning. In addition, ESM-enabled boundary spanning played a mediating role between sense of power and radical creativity.

5.4 Moderating effect

The PLS-product indicator approach in SmartPLS 3.0 was used to examine the moderating effects of boundary buffering (Zhao and Wagner, 2023). As shown in Table 7, the moderating effect of boundary buffering on the relationship between supervisor-subordinate instrumental ties on ESM and sense of power was not significant ($\beta = -0.059, t = 1.197$), indicating that H6b was not supported. However, boundary buffering was found to strengthen the positive relationship between supervisor-subordinate expressive ties on ESM and sense of power

Table 6. Mediation test

Path	Effect	t-statistics	p-values	Lower CI	Upper CI	Decision
SSET → SP → RC	0.262	6.253	0.000	0.181	0.338	Supported
SSET → SP → BS	0.329	7.057	0.000	0.236	0.412	Supported
SSIT → SP → BS	0.097	2.689	0.007	0.025	0.173	Supported
SSIT → SP → RC	0.077	2.715	0.007	0.022	0.133	Supported
SP → BS → RC	0.209	4.217	0.000	0.120	0.317	Supported

Table 7. Moderation test

No	Hypothesized relationship	Path coefficient	t-statistics	p-values	Lower CI	Upper CI	Decision
H6a	SSET × BB → SP	0.128	2.772	0.006	0.045	0.225	Supported
H6b	SSIT × BB → SP	−0.059	1.294	0.196	−0.166	0.013	Not supported

($\beta = 0.128$, $t = 2.604$), indicating that H6a was supported. This moderating effect was illustrated in Figure 3.

5.5 Fuzzy-set qualitative comparative analyses

This study employed fsQCA to conduct an in-depth examination of the data and identify combinations of antecedent conditions that lead to high levels of radical creativity. The fsQCA procedure consisted of three main steps: calibration, necessary conditions analysis, and sufficient conditions analysis, as described below.

5.5.1 Calibration. In the first step, fuzzy-set calibration was conducted to transform raw data into set membership scores ranging from 0 (full non-membership) to 1 (full membership), following the principles of set-theoretic methods (Ragin, 2008). Consistent with fsQCA methodological standards, this study used fsQCA 4.0 to perform the calibration. Based on the approach proposed by Ragin and Strand (2008), three qualitative anchors were defined: full membership (fuzzy score = 0.95), full non-membership (fuzzy score = 0.05), and the crossover point (fuzzy score = 0.5).

5.5.2 Analysis of necessary conditions. Before examining sufficient configurations, this study used fsQCA 4.0 to test whether any single condition was necessary for achieving high radical creativity. The five conditions analyzed were: supervisor-subordinate instrumental ties on ESM, supervisor-subordinate expressive ties on ESM, boundary spanning, sense of power, and boundary buffering. The consistency scores for all conditions ranged from 0.474 to 0.857, which are below the standard threshold of 0.90 (Ragin, 2008). This result indicated that no individual condition was sufficient on its own to produce high radical creativity. Instead, radical creativity emerged from the combined influence of multiple interacting factors. These findings provided the basis for proceeding to the analysis of sufficient configurations using fsQCA.

5.5.3 Analysis of sufficient conditions. Following the necessity analysis, this study employed fsQCA 4.0 to identify the sufficient conditions for achieving high levels of radical creativity. Given that the sample size exceeded 300 observations, the minimum frequency threshold for constructing the truth table was set at 3, and the raw consistency cutoff was set at

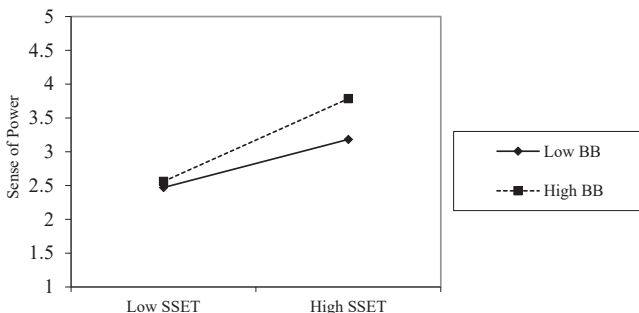


Figure 3. Moderating role of boundary buffering

0.85 (Pappas and Woodside, 2021). The analysis focused on the intermediate solution, which balances complexity and interpretability. To test the robustness of the results, the consistency threshold was lowered to 0.80. The core configurations remained stable, confirming the reliability of the findings (Qi *et al.*, 2025b).

As shown in Table 8, the fsQCA results revealed three distinct pathways for achieving high levels of radical creativity, each demonstrating a consistency score exceeding 0.8. The first configuration indicated that a combination of high supervisor-subordinate instrumental ties, sense of power, boundary spanning, and boundary buffering could still result in high radical creativity. The second configuration suggested that the presence of high supervisor-subordinate expressive ties, sense of power, boundary spanning, and boundary buffering could still foster high levels of radical creativity. The third configuration revealed that even with low levels of boundary buffering, a combination of high supervisor-subordinate instrumental ties, supervisor-subordinate expressive ties, sense of power, and boundary spanning could still lead to high radical creativity. These findings highlighted the principle of equifinality, illustrating that multiple pathways could lead to similar outcomes in terms of radical creativity, reflecting the complex and nonlinear nature of these relationships.

The results indicated that sense of power and boundary spanning were key conditions leading to high levels of radical creativity. They also revealed an interdependent and complementary relationship among supervisor-subordinate expressive ties, supervisor-subordinate instrumental ties, and boundary buffering in achieving high levels of radical creativity. When one of these factors was absent, the presence and reinforcement of the other two were crucial for sustaining high levels of radical creativity.

6. Discussions

6.1 Key findings

First, the results show that both instrumental and expressive supervisor-subordinate ties on ESM significantly enhance employees’ sense of power. This finding not only supports AITP’s proposition that the sense of power arises from individuals’ perceived control over valuable resources in social interactions (Anderson *et al.*, 2012) but also extends the theory by revealing that ESM-mediated supervisor-subordinate ties amplify employees’ sense of power through greater perceived access to resources and recognition from supervisors. Despite previous studies indicating that leadership traits and style influence sense of power (Luo and Chen, 2024; Nguyen *et al.*, 2024), this study further clarifies how different types of supervisor-subordinate ties affect employees’ sense of power in ESM contexts. Specifically, the results show that supervisor-subordinate expressive ties on ESM have a more substantial influence on

Table 8. Configurations for achieving high levels of radical creativity

Configuration	Solution 1	2	3
SSET		●	●
SSIT	●		●
SP	●	●	●
BS	●	●	●
BB	●	●	
Consistency	0.967	0.951	0.965
Raw coverage	0.619	0.649	0.623
Unique coverage	0.032	0.062	0.035
Solution consistency	0.717		
Solution coverage	0.942		

Note(s): Black circles (●) indicate the presence of a condition. Blank spaces indicate “don’t care”

sense of power than instrumental ties. One possible explanation lies in the Chinese cultural context, where employees tend to view emotional closeness and informal support from supervisors as signals of personal trust and recognition, which in turn enhance their perceived control and empowerment (Wang *et al.*, 2019; Khan *et al.*, 2023). These findings offer practical guidance for supervisors on how to use ESM to foster employees' psychological empowerment in digitally mediated workplaces.

Second, the results show that employees' sense of power has a positive effect on radical creativity, with ESM-enabled boundary spanning serving as a mediating mechanism. This finding not only supports AITP's proposition that a high sense of power activates the approach system (Keltner *et al.*, 2003; Luo *et al.*, 2024) but also extends the theory by revealing that sense of power serves as an approach-oriented psychological condition that encourages employees' proactive use of technology. Despite previous studies in traditional workplaces having identified the sense of power as a key antecedent of proactive behaviors such as taking charge behaviors and error risk-taking (Luo *et al.*, 2024; Song and Gu, 2024b), this study further indicates that sense of power motivates employees to proactively use ESM to span boundaries for diverse and heterogeneous resources, fostering radical creativity. The fsQCA results also reveal that the sense of power is a core condition across multiple pathways leading to high radical creativity. Building on previous studies that highlight ESM's potential to support creativity through its affordances and use purposes (Fu *et al.*, 2024; Zhang *et al.*, 2024b), this study further highlights sense of power as the central psychological mechanism through which supervisor-subordinate ties on ESM activate that potential. These findings underscore the importance of cultivating employees' sense of power to fully realize ESM's creative benefits.

Finally, the results show that boundary buffering significantly strengthens the positive effect of supervisor-subordinate expressive ties on employees' sense of power in ESM contexts. As hypothesized, boundary buffering enhances employees' perceived control within these expressive ties, enabling them to access resources and support from supervisors more effectively. This finding supports the core proposition of AITP that sense of power arises from individuals' perceived control over valuable resources rather than from formal authority (Anderson *et al.*, 2012; Ni and Zheng, 2024). It further suggests that boundary buffering can amplify employees' sense of power in ESM-based interactions. However, the moderating effect of boundary buffering on the relationship between instrumental ties and sense of power is not significant. A possible explanation lies in the nature of instrumental ties, which are task-oriented and clearly structured (Pitafi and Kiani, 2025; Zhang *et al.*, 2025b), leaving limited scope for boundary buffering to influence perceived control. In contrast, expressive ties involve more informal, emotionally charged communication, which can lead to social overload and interruptions. In such contexts, boundary buffering helps maintain psychological autonomy and safety (Faraj and Yan, 2009; Zhang *et al.*, 2022). These nuanced findings highlight the importance of fostering employees' boundary buffering capabilities to fully leverage ESM for psychological empowerment and creativity.

6.2 Theoretical implications

This study makes several theoretical contributions. First, it contributes to the literature on ESM use and radical creativity by revealing the relationship between supervisor-subordinate ties on ESM and radical creativity. Existing research primarily explores the functional potential of ESM to support employee creativity based on its features and use (Fu *et al.*, 2024; Zhang *et al.*, 2024b; Shao *et al.*, 2025b), but offers limited insights into how supervisors can implement interventions to realize this potential. Although recent studies identify leadership as a critical driver of radical creativity (Song *et al.*, 2025; Zhu *et al.*, 2025c), it remains unclear how these effects unfold when ESM transforms communication and interaction patterns between supervisors and subordinates (Budjanovcanin and Yang, 2025; Ollier-Malaterre and Redston, 2025). This study fills this research gap by examining how different types of supervisor-

subordinate ties on ESM influence radical creativity. The results show that both instrumental and expressive ties enhance employees' sense of power, which in turn promotes ESM-enabled boundary spanning and stimulates radical creativity. Extending prior work that highlights ESM's potential to foster creativity through horizontal collaboration among employees, this study demonstrates that vertical supervisor-subordinate ties play a pivotal role in unlocking this potential. These findings suggest that organizations should not only focus on deploying advanced ESM functionalities but also encourage supervisors to actively leverage ESM to engage in diverse interactions with subordinates, thereby fostering employees' proactive and effective use of ESM.

Second, it contributes to the literature on supervisor-subordinate ties and AITP by revealing the mediating role of sense of power in linking supervisor-subordinate ties on ESM to radical creativity. Although recent qualitative studies indicate that ESM supports more direct ties between supervisors and subordinates to influence employee outcomes (Budjanovcanin and Yang, 2025), there remains a lack of models based on theoretical and empirical evidence to explain whether and why supervisor-subordinate ties on ESM can foster radical creativity. AITP emphasizes that the approach-effect driven by sense of power is crucial for individuals to engage in proactive behavior (Luo *et al.*, 2024; Wang *et al.*, 2025), but how sense of power functions in ESM contexts remains unexplored, although research indicates that ESM affordances enhance employees' psychological empowerment (Sun *et al.*, 2024; Khan *et al.*, 2025). This study fills these gaps by revealing that sense of power mediates the relationship between supervisor-subordinate ties on ESM and radical creativity. This study offers a novel perspective on understanding the underlying mechanisms of supervisor-subordinate ties and radical creativity. By revealing the antecedents and consequences of sense of power in ESM contexts, this study verifies AITP's applicability in ESM contexts and expands its explanatory power in the digital workplace. These findings suggest that managers should view ESM not merely as a monitoring tool, but rather actively leverage its diverse functionalities to empower employees, thereby unlocking ESM's potential to drive creativity.

Finally, it contributes to the literature on ESM use and sense of power by revealing the moderating role of boundary buffering in the relationship between supervisor-subordinate ties and sense of power. Previous studies have mainly explored the boundary conditions of external factors influencing employees' sense of power from the perspective of individual characteristics (e.g. individual identity orientation) and organizational characteristics (e.g. teamwork atmosphere) (Ni and Zheng, 2024; Zhang *et al.*, 2024a), offering limited insight into how employees can actively shape and maintain their sense of power in digitally mediated interactions. To address this gap, this study, drawing on AITP, which posits that sense of power arises from perceived control over valuable social resources (Wang *et al.*, 2025), finds that employees with boundary buffering capabilities can amplify the positive influence of supervisor-subordinate expressive ties on sense of power by maintaining focus and autonomy. By highlighting the moderating role of boundary buffering, this study further enhances our comprehension of the conditions that impact the development of sense of power in ESM contexts. These findings suggest that organizations should not only encourage employees to use ESM for boundary-spanning activities that enhance resource access, but also help them develop boundary buffering skills to balance open communication with psychological autonomy.

6.3 Practical implications

This study offers several managerial implications. First, this study provides practical insights for managers on how to enhance radical creativity in ESM contexts. The results show that supervisor-subordinate ties on ESM positively influence boundary spanning and radical creativity through employees' sense of power. Therefore, organizations should encourage and guide supervisors to use ESM not only to build instrumental ties but also to develop expressive ties that foster creativity. Specifically, organizations should cultivate a supportive, open, and

egalitarian digital communication climate, encouraging supervisors and subordinates to engage in both work-related and non-work-related interactions on enterprise social media, thereby reducing traditional hierarchical constraints and promoting freer social connections. Furthermore, organizations should offer targeted training programs for supervisors to help them develop specific skills for establishing high-quality supervisor-subordinate ties on ESM. For example, organizations can train supervisors to utilize ESM features (e.g. dynamic information flow, tags, task updates) to continuously monitor employee progress and provide timely task guidance (Ollier-Malaterre and Redston, 2025), thereby establishing efficient instrumental ties. Meanwhile, organizations should also train supervisors to use ESM features, such as comments, emojis, and likes, to balance instrumental feedback with emotional support, thereby fostering high-quality expressive ties. By cultivating a positive digital communication climate and improving supervisors' ESM competencies, supervisors can more effectively utilize ESM to establish diverse ties with their subordinates, thereby stimulating employees' radical creativity.

Second, the results highlight the mediating role of employees' sense of power in the relationship between supervisor-subordinate ties and radical creativity, underscoring the importance of empowering employees in ESM contexts. To encourage boundary-spanning behaviors and stimulate radical creativity, supervisors should use ESM to enhance employees' perceived control and influence over their work. For example, supervisors can utilize the ESM feature to invite employees to recommend task priorities and publicly recognize their solutions, helping them feel empowered and in control of the workflow. Notably, the results show that expressive ties have a stronger effect on employees' sense of power than instrumental ties, indicating that affirmation and emotional support are particularly effective for fostering psychological empowerment in ESM contexts. Therefore, supervisors should focus on building expressive ties by using ESM to convey appreciation and emotional warmth, such as through personalized messages or informal check-ins that make employees feel valued. In addition, organizations should establish clear ESM use policies to guide supervisors in managing the frequency and timing of interactions, preventing social overload and interruptions (Luqman *et al.*, 2021; Sun *et al.*, 2025), which can undermine employees' autonomy. These practices can help supervisors use ESM more effectively to empower employees and enhance radical creativity.

Finally, the results show that boundary buffering strengthens the positive effect of supervisor-subordinate expressive ties on employees' sense of power in ESM contexts, suggesting that employees are more likely to feel empowered and in control when they can effectively manage their digital boundaries during social interactions. To support this, organizations should tailor the design of ESM platforms to balance the promotion of constructive social ties with effective boundary management. On one hand, the design of ESM platforms should incorporate features such as "interaction frequency reminders," "non-urgent message labels," and "appropriate communication prompts" to prevent expressive ties from escalating into high-pressure, excessive communication. On the other hand, features such as "do-not-disturb modes," "adjustable online visibility," and "message prioritization filters" (Zhang *et al.*, 2023) should be provided to allow employees to manage their digital boundaries and maintain control over communication rhythms. In addition, organizations should provide training on digital boundary management to help both supervisors and employees maintain open yet respectful communication practices. By integrating supportive technological features with organizational policies, organizations can reduce digital overload while facilitating high-quality supervisor-subordinate ties, maximizing ESM's potential to empower employees and enhance creativity.

6.4 Limitations and future research

This study has several limitations that provide opportunities for future research. First, this study only explored how supervisor-subordinate ties on ESM influence ESM's support for

radical creativity from the perspective of vertical social relationships. Future research could broaden this perspective by examining how peer ties on ESM affect ESM use. Second, this study investigated the impact of different supervisor-subordinate ties on employees' sense of power in the digital workplace. Given the diverse sources of sense of power, future research could explore how to enhance employees' sense of power in digital workplaces from different perspectives, such as technology affordances. Third, this study relied on cross-sectional data collected through a questionnaire; future research could validate the findings using longitudinal data collection. Finally, this study was conducted in China, and considering that sense of power is subjective, it may differ across cultures (To *et al.*, 2020). Future research could examine how cultural differences influence the results.

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